

# Evolutionary Algorithm

## What Is It?

An **Evolutionary Algorithm (EA)** is an optimization method inspired by biological evolution.

Instead of starting with one solution and improving it step by step using gradients, an evolutionary algorithm maintains a **population of candidate solutions**, evaluates them, keeps the better ones, modifies them, and repeats.

## A Simple Example

Suppose we want to find the best prompt for an LLM. We begin with several candidate prompts:

```
Population:
P1 = "Solve the problem carefully."
P2 = "Think step by step and verify your answer."
P3 = "Analyze the problem, solve it, then check for mistakes."
P4 = "Act as an expert mathematician."
```

We evaluate each prompt on some tasks:

```
P1 → 65% accuracy
P2 → 82% accuracy
P3 → 88% accuracy ★ best
P4 → 73% accuracy
```

Then the evolutionary algorithm performs something like the steps below.

### 1. Selection

Keep the good solutions

P2, P3

### 2. Mutation

Slightly modify a good solution

P3: "Analyze the problem, solve it, then check for mistakes."

⇓ *mutation*

"Analyze the problem step by step, verify intermediate results, then check the final answer."

### 3. Crossover (Optional)

Combine parts of two good solutions

P2 + P3

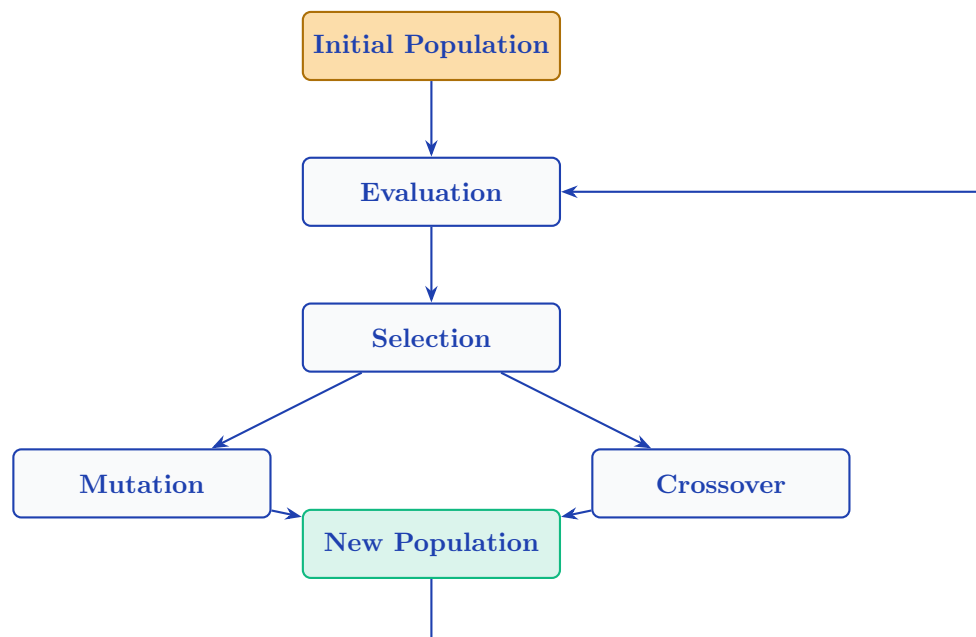


"Think step by step, solve the problem, and verify your final answer."

### 4. Evaluation

Test the new candidates.

The process then repeats. Conceptually:



## The Fitness Function

The objective being measured is usually called the **fitness function**. For example,

$$\text{Fitness}(P_i) = \text{Accuracy of prompt } P_i$$

and the algorithm tries to find

$$P^* = \arg \max_{P_i} \text{Fitness}(P_i).$$

## Why It Matters for Text

Evolutionary algorithms are particularly interesting when the thing you're optimizing is **discrete or textual** — for example, prompts, agent workflows, tool-use strategies, or skills.

You cannot easily calculate a gradient like

$$\frac{\partial \text{Accuracy}}{\partial \text{Prompt}}$$

because a prompt is text rather than a continuous parameter vector.

So instead, you cycle through:

